Roll Number | | | | | | | | | | | | |

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
EE(CS)201 - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

TIME ALLOTTED: 3Hours **Maximum Marks : 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Part A
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.	In an AC circuit, what is the phase relationship between voltage and current in a purely resistive element?	**(1)**	CO2	BL4	PO1
	(a) Voltage leads current by 90 degrees (b) Voltage lags current by 90 degrees (c) Voltage and current are in phase (d) Voltage and current are out of phase by 180 degrees				
2.	In a DC motor-generator set, what is the primary function of the DC motor?	**(1)**	CO1	BL3	PO2
	(a) To convert DC to AC (b) To convert mechanical energy to DC (c) To provide electrical power to the load (d) To regulate voltage in the system				
3.	A semiconductor is a material whose conductivity is:	**(1)**	CO3	BL2	PO2
	(a) Equal to a conductor (b) Equal to an insulator (c) Between a conductor and an insulator (d) Zero				
4.	The most commonly used OP-AMP IC is:	**(1)**	CO3	BL1	PO1
	(a) 555 (b) 741 (c) 8085 (d) 8051				

5.	In an intrinsic semiconductor, the number of electrons is:	**(1)**	CO3	BL2	PO1

(a) Greater than holes (b) Less than holes

(c) Equal to holes (d) Zero

6.	A transistor has _____	**(1)**	CO4	BL1	PO1

(a) Four PN junction (b) One PN junction

(c) Three PN junction (d) Two PN junction

7.	How many valence electrons are there in semiconductors?	**(1)**	CO3	BL2	PO1

(a) 3 (b) 1

(c) 2 (d) 4

8.	The gain for an ideal non-inverting operational amplifier is (given R_2 is the feedback resistance)	**(1)**	CO4	BL2	PO3

(a) R_2/R_1 (b) $-R_2/R_1$

(c) $(R_2/R_1 + 1)$ (d) $(R_2/R_1 + 1)$

9.	What is the power factor in a pure resistive circuit?	**(1)**	CO2	BL1	PO1

(a) -1 (b) 0

(c) 1 (d) Infinity

10.	Identify the main function of a BJT?	**(1)**	CO3	BL2	PO2

(a) Acts as amplifier (b) Acts as a switch

(c) Acts as a rectifier (d) Both a and b

11.	In any A.C. circuit always	**(1)**	CO1	BL1	PO1

(a) Apparent power is more than actual power (b) Reactive power is more than apparent power

(c) Actual power is more than reactive power (d) Reactive power is more than actual power

12.	A semiconductor material has a band gap of approximately	**(1)**	CO4	BL2	PO1

(a) 0 eV (b) 1 eV

(c) 10 eV (d) 100 eV

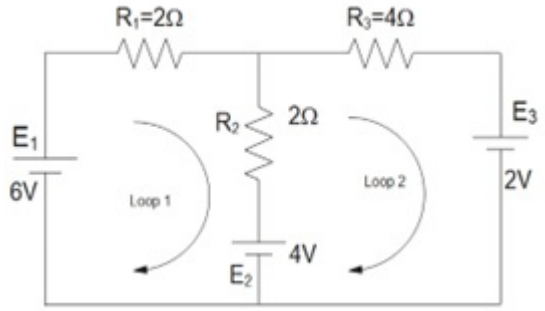
Part B
(Answer any three of the following) 3x5=15

13. (a)	Establish a relationship of gain of the amplifier with feedback & without feedback.	**(2)**	CO3	BL4	PO3

(b)	With a negative feedback, an amplifier gives an output of 10V with an input of 0.5 V. When the feedback is removed, it requires 0.25 V input for the same output. Calculate (i) Gain without feedback (ii) feedback fraction β.	**(3)**	CO3	BL4	PO3
14.	Explain the construction and working of a BJT.	**(5)**	CO4	BL2	PO3
15.	Draw the circuit diagram of close loop non- inverting amplifier and derive expression of its voltage gain.	**(5)**	CO4	BL3	PO2
16.	Derive the RMS value of alternating current.	**(5)**	CO1	BL2	PO2
17.	Derive the equation for the transformation ratio in a transformer.	**(5)**	CO2	BL1	PO2

Part C
(Answer any three of the following) 3x15=45

18. (a)	Compare conductors, semiconductors, and insulators	**(5)**	CO3	BL4	PO2
(b)	Define a semiconductor. Explain its characteristics with examples.	**(5)**	CO3	BL1	PO2
(c)	What is an intrinsic semiconductor? Explain its properties.	**(5)**	CO3	BL2	PO2
19. (a)	Draw circuit diagram of clipper circuit and explain it.	**(5)**	CO2	BL2	PO2
(b)	Explain different feedback topologies	**(5)**	CO2	BL2	PO2
(c)	Draw circuit diagram of Integrator circuit and find output expression.	**(5)**	CO3	BL2	PO2
20. (a)	Draw and Explain I-V characteristics of BJT.	**(5)**	CO3	BL2	PO2
(b)	Using KVL find the branch currents (I1 &I2) in the given circuit.	**(10)**	CO1	BL3	PO3



21. (a)	Draw and explain the V-I characteristics of a Zener diode.	**(6)**	CO3	BL4	PO2

(b)	Explain how a Zener diode is used as a voltage regulator.	**(6)**	CO3	BL2	PO2
(c)	Describe the differences between an ideal diode and a real diode.	**(3)**	CO3	BL2	PO2
22.	Draw the pin configuration and equivalent circuit of an Operational Amplifier (IC 741) and explain its working.	**(15)**	CO4	BL3	PO1